Edge Cases

Team work great = user error

Pathfinding/movement:

1. Gantry loses track of its current position.
2. User tries to move the gantry outside of its allowed workspace.
3. Two movement inputs are given at the same time/conflicting commands are received.
4. Movement commanded before homing.
5. Actual position does not match its expected position
6. Pathfinding can't connect start and end points
7. Remote pathfinding connection is broken during move

Safety:

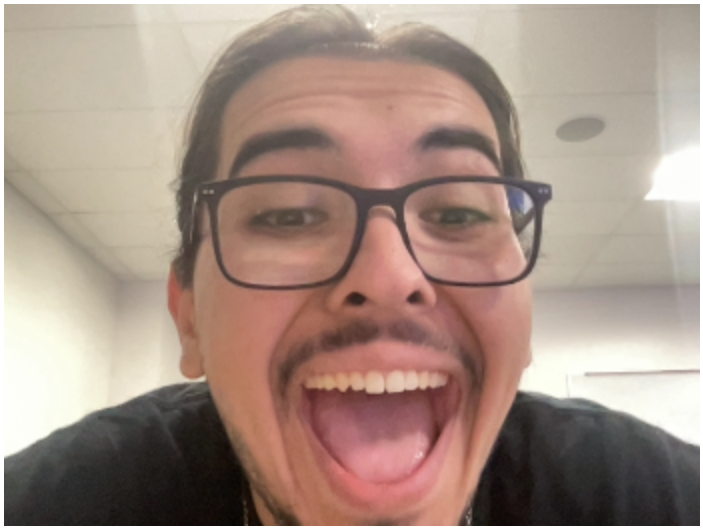
1. Ball of the wrecking swinging out (load swings too far during or after movement)
2. Weight Scale Sensor, giving input for less weight than what is attached
3. Component catches fire.
4. Lack of power delivery or too little power
5. Emergency stop is pressed while the gantry is moving a load.
6. Motor stalls while the gantry is moving.
7. Power goes out while a load is suspended.
8. Hook or payload becomes disconnected while moving.
9. Load exceeds the limit after it's already lifted.
10. Power is lost while the gantry is moving horizontally
11. Limit switch triggered unexpectedly.
12. Weight sensor fails completely or stops sending data.
13. User tries to lift or lower while the gantry is moving.

Attachment:

1. Attachment is not fully secured before the gantry starts moving.
2. Attachment becomes loose or disconnects during movement.
3. Wrong attachment is installed for the selected task.
4. System fails to detect the installed attachment.
5. Attachment is too heavy for the gantry.
6. Attachment collides with the frame or another object.
7. Attachment stops working during use.
8. User tries to operate an attachment that is not installed.

Later:

1. User walks into payload



Goofy asl